

Maintenance Manual for Elevator Door Operators

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1. Introduction

The scope of this maintenance manual covers the main components of elevator door operators that may greatly compromise the safety of users and cause serious accidents if malfunctions or insufficiently managed.

Elevator door operators need to be inspected and maintained periodically by competent maintenance persons. If used without proper maintenance, they may not be able to deliver the performance we expect. The components specified in this manual are particularly critical for safety of users and maintenance persons. Please formulate the appropriate maintenance plan in accordance with this manual.

For maintenance of Mitsubishi Electric elevators, we recommend that you sign a maintenance contract with our official distributor. To contact our official distributor, please visit our website:

www.mitsubishielectric.com/elevator/network/index.html.

This manual describes important items that need to be checked with particular care in performing basic maintenance.

The owner of the installation and operation manager must request the maintenance organization to perform maintenance in accordance with this manual.

2. Maintenance of door operator

The main components of elevator door operators described in this manual refer to the critical parts that may cause serious accidents involving users if their functions and states are improperly maintained. Because the way in which the state of the elevator changes varies depending on the operation time and installation environment, the maintenance activity must be performed to maintain the appropriate state at all times.

This document describes items that must be checked in performing basic maintenance for the main components of elevator door operators mentioned above.

3. General precautions

This manual summarizes important maintenance information for competent maintenance persons who carry out basic elevator maintenance. The competent maintenance persons shall understand and observe the instructions thoroughly.

3.1 Safety symbols

Safety symbols below represent the degree of hazard that would arise should the provided instructions be neglected. The definitions of the symbols are as follows.

(1) Definitions of danger, warning, and caution symbols

Symbol	Description
 Danger	Indicates an imminently hazardous situation which, if not observed, will result in death or serious injury.
 Warning	Indicates a potentially hazardous situation which, if not observed, could result in death or serious injury.
 Caution	Indicates a potentially hazardous situation which, if not observed, may result in injury or damage to the elevator equipment.

(2) Definitions of precaution symbols

Symbol	Description
	Indicates a mandatory action.
	Indicates a prohibited action.
	Warns of electricity. (This symbol reminds workers to take care to avoid coming into contact with live-wire portions.)

3.2 Precautions after inspection and maintenance

(1) Fault

If any fault has been found during inspection and/or maintenance, take appropriate measures immediately.

- 1) Take the elevator out of service until the fault is repaired. Report on the state to the operation manager.
- 2) Record the details of fault, replacement, and repair in the Work Log, and maintain it permanently.
- 3) If any abnormality has been found during inspection, and replacement, repair or adjustment by Mitsubishi Electric Corporation is required, please contact our official distributor.

 Caution	 While the elevator is out-of-service, be sure to take measures to prevent anyone from entering the area.  Use the Mitsubishi Electric's genuine parts for replacement.  If there is any problem with the product, contact our official distributor.
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(2) Restoration

- 1) Restore any screws, covers, or other parts loosened or removed during inspection and/or maintenance to the original state.
- 2) If no abnormality has been found during inspection and/or maintenance, confirm the safety and restore the elevator to the automatic (normal) operation.

 Caution	 Before resuming the automatic (normal) operation, check that there is no problem with elevator operation by first running the car manually, followed by stopping the car at every floor in automatic (normal) operation, and then running the car from the top to the bottom floor in automatic (normal) operation at the end.
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3.3 Service parts

For supply of parts, contact our official distributor.

Please note that we may not be able to supply some parts depending on when the product was produced and the condition of the equipment. In such case, we recommend modernization of your product.

 Caution	 This manual is subject to change without notice. Before starting maintenance, visit the URL below to check the latest manual. www.mitsubishielectric.com/elevator/maintenance/index.html
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4. Inspection items and standard intervals

Follow the table below to check the state of each part. If any abnormality is found as the result of the inspection, replace the part(s) concerned.

	Item	Description	Interval (standard)
Car door operator	Belt and pulley	- Check that the fixed portions of belts and pulleys are not loose. [Details 5. (1)] - Check the belt tension and how the belt is wound around the pulley. [Details 5. (2) (4)] - Check that there are no cracks, wear and other abnormalities on the belt. [Details 5. (3) (5)] - Check that there is no looseness on the crimped part of pulley shaft, and there are no looseness and rotation failure, wear and rust on the bearing. Make sure to check the pulley itself (without applying tension) when replacing the belt (Recommended replacement interval: Approximately 12 years).	12 months
	Link	- Check that the individual bolts are not loose. - Check that there are no significant looseness, wear and rust on the shaft and bearing portions. - Check that alignment is properly adjusted. [Details 5. (11)]	12 months
	Interlocking chain	- Check that the individual bolts are not loose. - Check the chain tension. [Details 5. (6)] - Check that there are no elongation, cracks, kinks and other abnormalities on the chain. - Check that there are no significant looseness, wear and rust on the shaft and bearing portions.	12 months
	Interlocking rope and pulley	- Check that the individual bolts are not loose. - Check the rope tension. [Details 5. (7)] - Check that there are no wire breakages, wear, rust and kinks on the entire rope. If there is rust and wire breakage or there is severe rust, replace the rope with new one immediately. (Breakage can be checked easily by rubbing the surface with a piece of cloth.) - Check that there is no looseness on the crimped part of pulley shaft, and there are no looseness and rotation failure, wear and rust on the bearing.	12 months
	Door hanger	- Check that the individual bolts are not loose. - Check that there are no significant looseness, wear and rust on the shaft and bearing portions of the hanger rollers. - Check that the clearance between the upthrust roller and the rail is within the standard range. [Details 5. (8)]	6 to 24 months
	Door shoe	- Check that the individual bolts are not loose. - Check that there are no wear and deformation on the door shoes.	3 to 12 months
	Car-door close detection switch	- Check that the switch is attached and adjusted to be within the standard range. [Details 5. (9)]	6 to 12 months
		- Check that the elevator does not start in the door open state.	1 to 3 months
Landing door operator	Interlocking rope and pulley	Same as the car	12 to 24 months
	Link	- Check that the individual bolts are not loose. - Check that there are no significant looseness, wear and rust on the shaft and bearing portions.	12 to 24 months
	Door hanger	Same as the car	12 to 24 months
	Door shoe	Same as the car	6 to 12 months
	Door closer	- Check that the doors automatically close. <u>Weight type</u> - Check that the individual bolts are not loose. - The clearance between the stopper and the pulley needs to be 2.5 mm or less. <u>Link type</u> - Check that there are no significant looseness, wear and rust on the shaft and bearing portions. <u>Spirator type</u> - Check that the rope does not overlap when it is being wound around the drum. - Check that the mounting bolts are not loose. - Check that there is no oil leakage.	3 to 24 months
	Interlock	- Pull the fully closed doors two or three times in the opening direction and check that the doors remain shut completely. - Check that there are no significant looseness, wear and rust on the shaft and bearing portions. - Check that the gap and engaged area between the latch and the hook are in the specified values. [Details 5. (10)]	3 to 24 months
		- Check that the elevator does not start in the door open state.	1 to 3 months

 Danger	 Due to the risk of falling into the hoistway during the inspection, take the necessary safety measures before the operation.  Since there is a risk of equipment malfunction/damage or serious accident, other than when operating manually, shut off the power supply in advance as necessary.  Since there is a risk of equipment malfunction/damage or serious accident, change the AUTO/HAND (NORMAL/INSPECTION) switch of the car operating device on car top to the HAND (INSPECTION) position.
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4.1 Car door operator (belt-driven type, link-driven type, and chain-driven type)

*For the inspection items of each component number, see section 5.

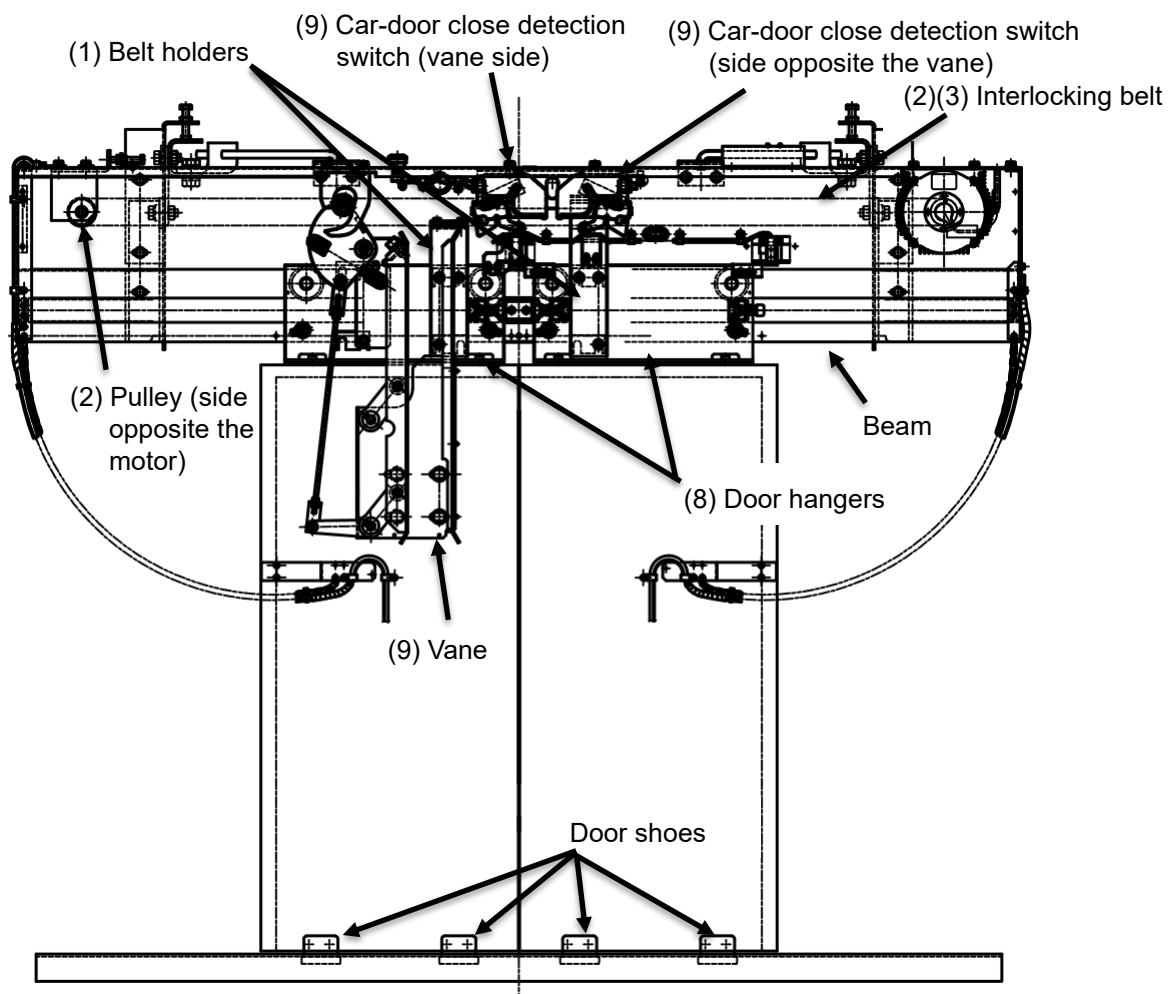


Fig. 4-1 Car door operator type 1 (belt-driven type)

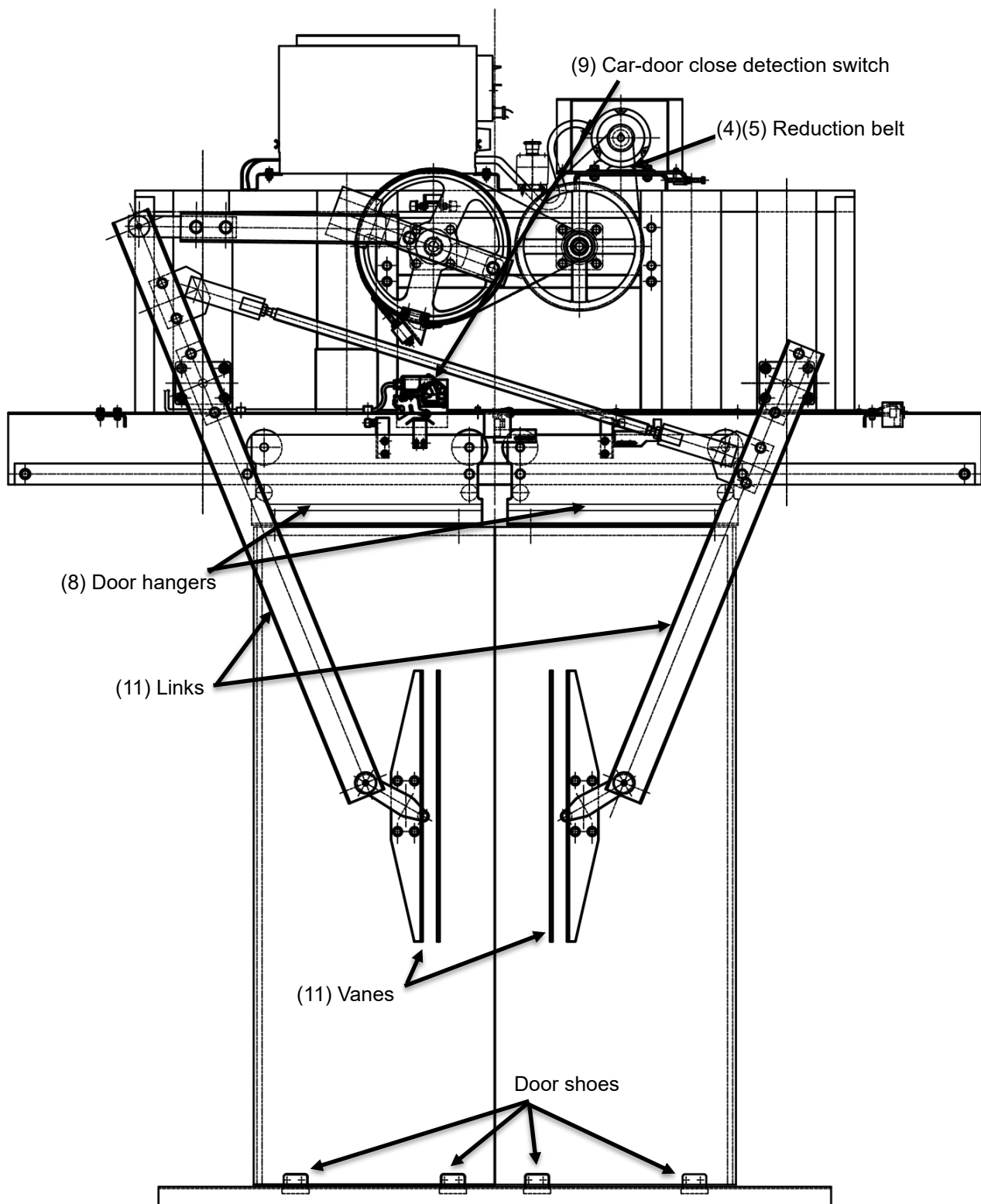


Fig. 4-2 Car door operator type 2 (link-driven type)

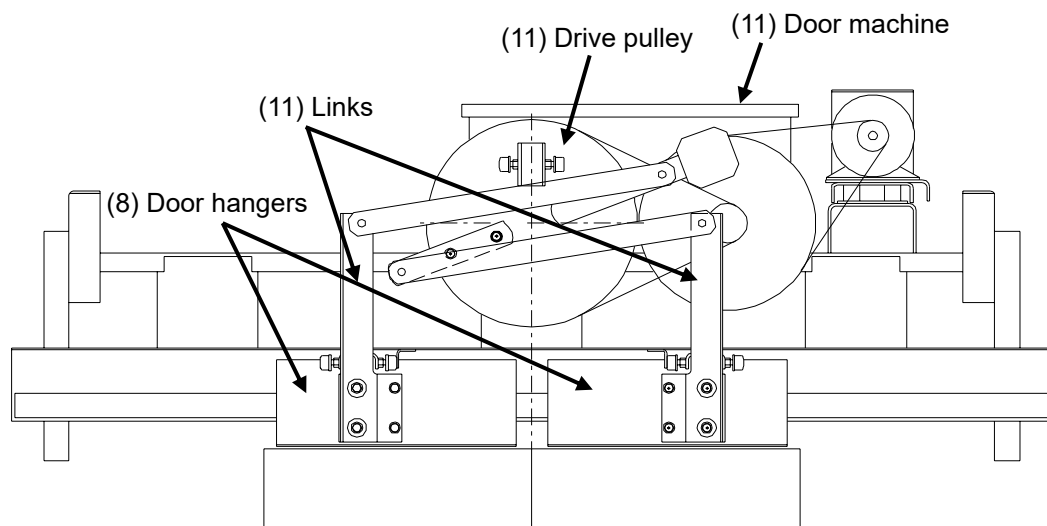


Fig. 4-3 Car door operator type 3 (double-link driven type)

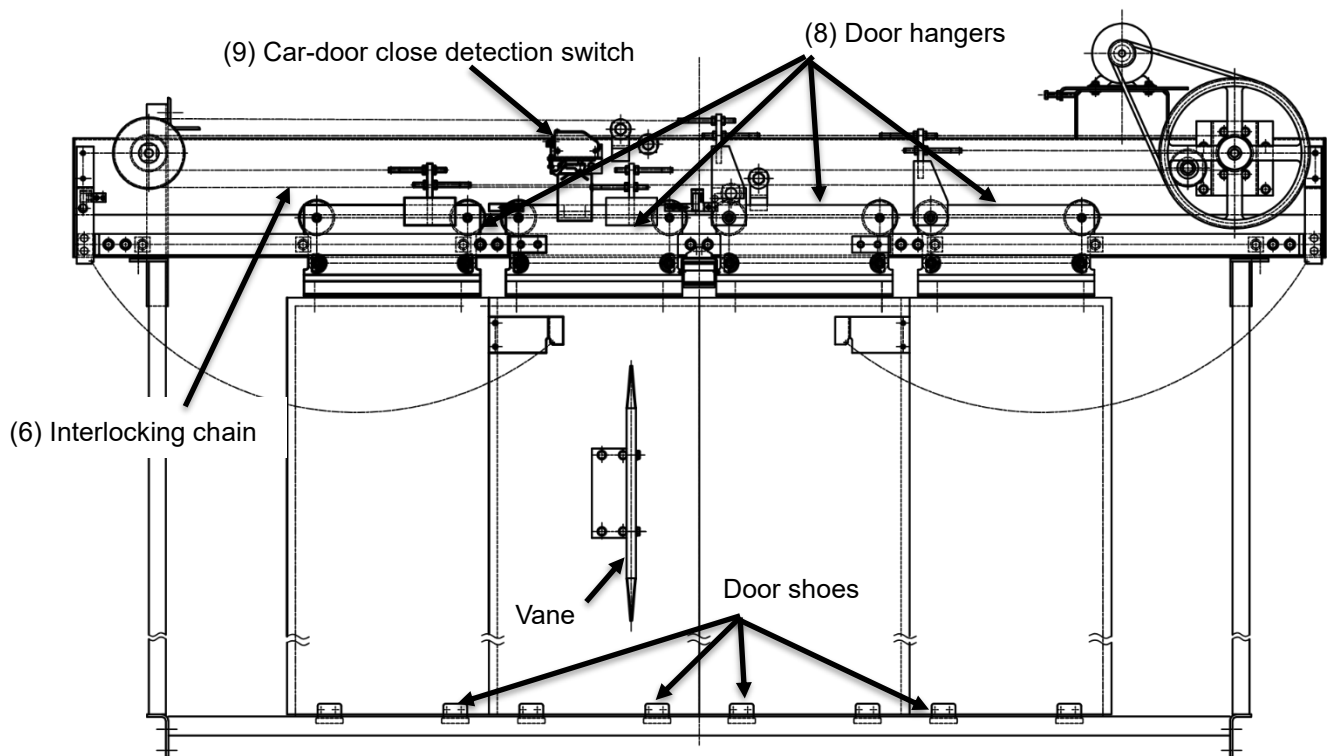


Fig. 4-4 Car door operator type 4 (chain-driven type)

4.2 Landing door operator (weight-closer type, link-closer type, and spirator type)

*For the inspection items of each component number, see section 5.

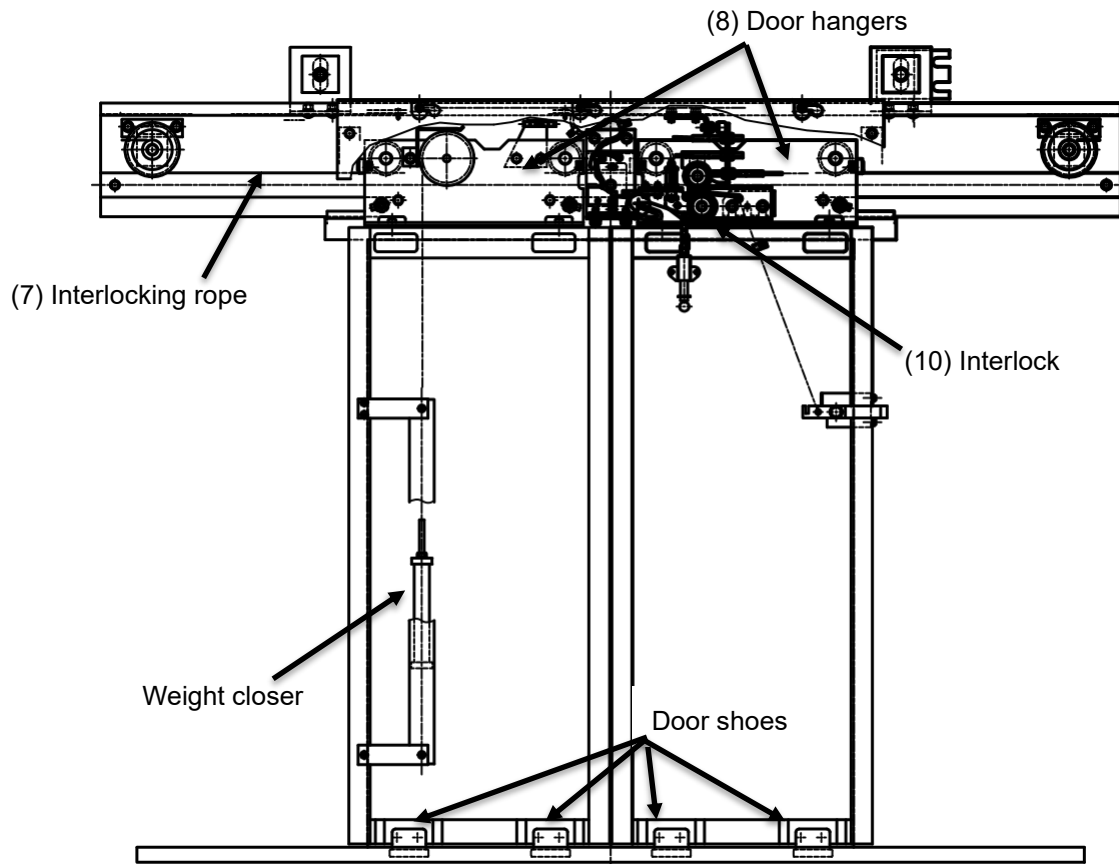


Fig. 4-5 Landing door operator type 1 (weight-closer type)

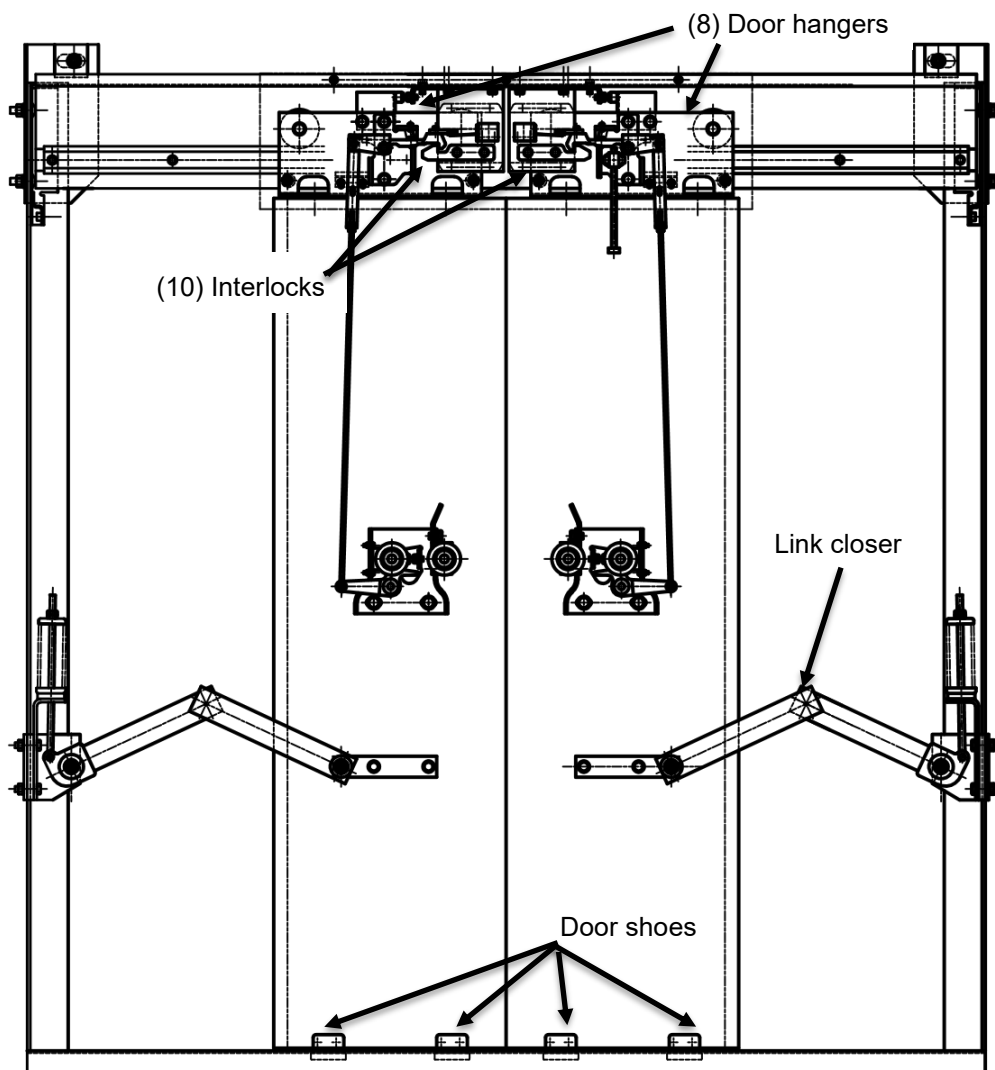


Fig. 4-6 Landing door operator type 2 (link-closer type)

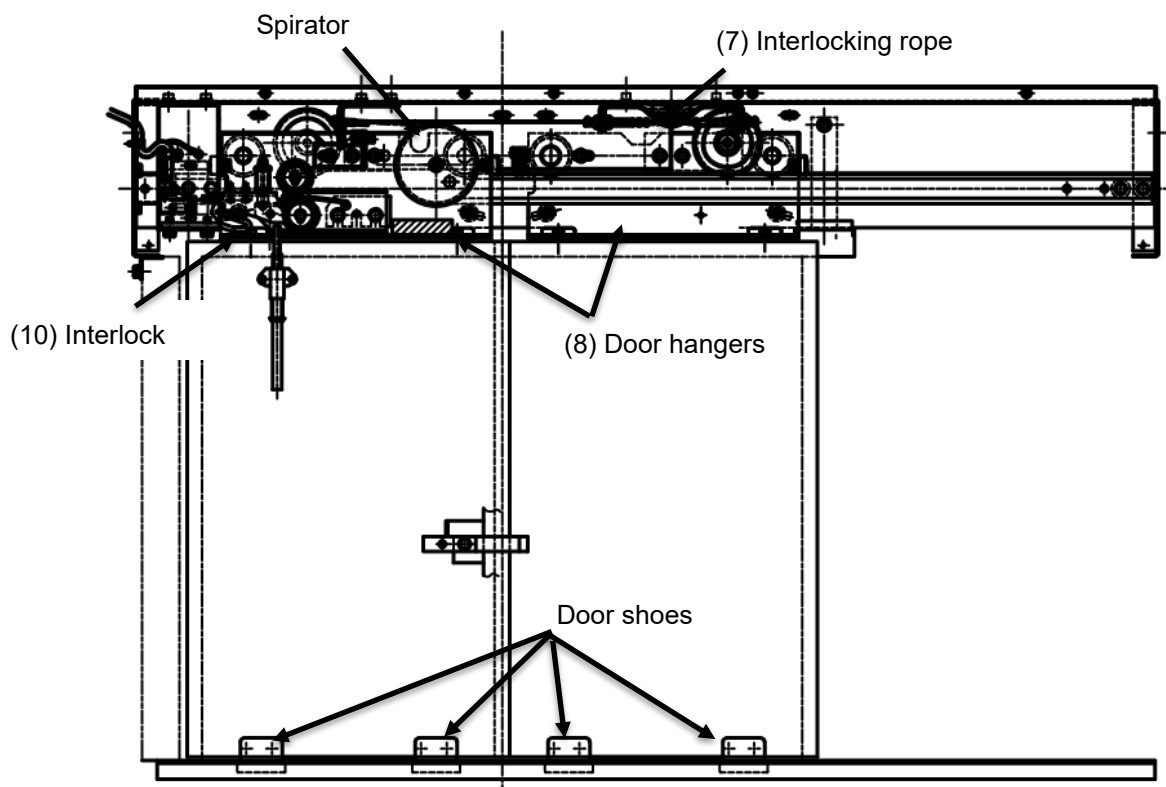


Fig. 4-7 Landing door operator type 3 (spirator type)

5. Details on inspection items

(1) Tightening torque for belt holder

The toothed belt is fixed with bolts (at belt holder) as shown in the figure below. Maintain the tightening torque of bolts within the range of 3 to 4 N·m.

*Check for retightening with a torque of 3 to 4 N·m at least every 12 months.

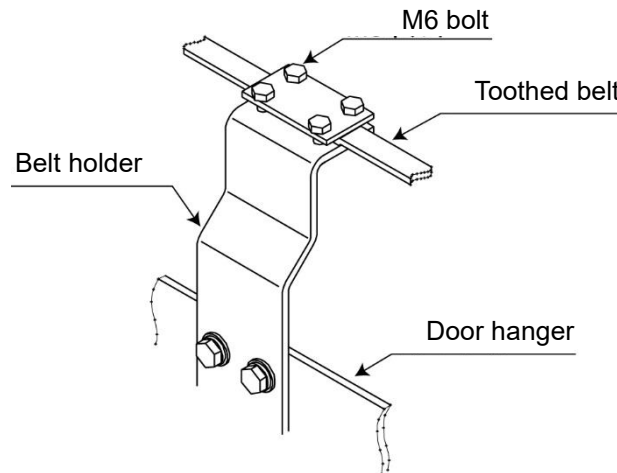


Fig. 5-1

(2) Tension and alignment of interlocking belt

(a) Check that the belt tension is maintained within the range below. Measure the tension at the center of the belt with the doors fully open.

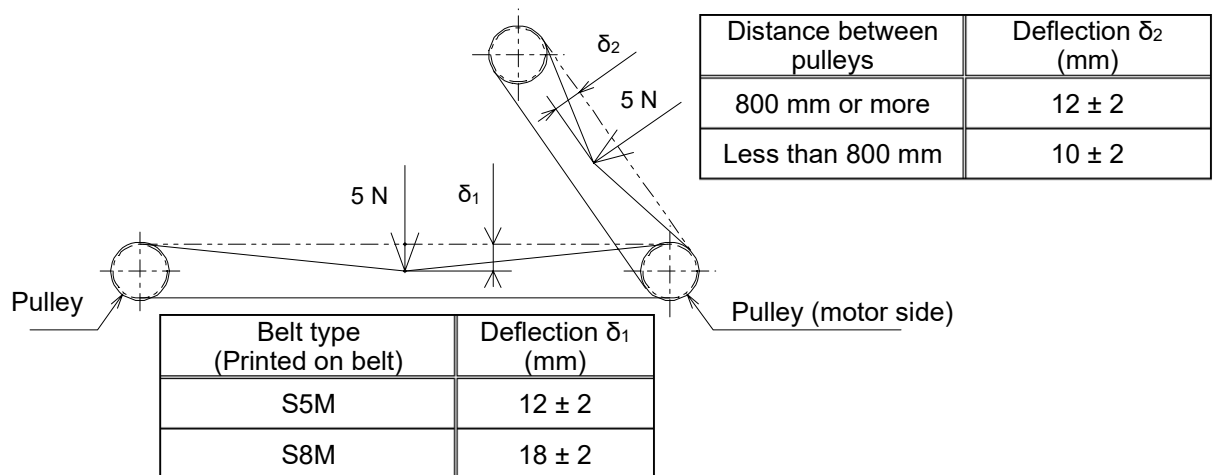


Fig. 5-2

- (b) Adjust the alignment using the fixing bolts on the pulley mounting plate so that the belt rotates straight along the pulleys. The adjustment is required especially when there is a squeaking noise in the belt and the flange and a large amount of abrasion powder is formed due to strong contact between the belt and the flange.

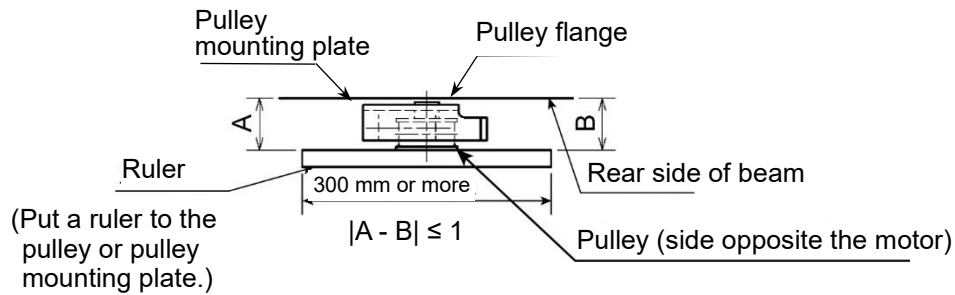


Fig. 5-3

(3) State of interlocking belt

Check that there are no cracks, wear, deformation and other abnormalities on the belt. Replace the belt if there is any crack or wear on the belt as shown in the figure below.

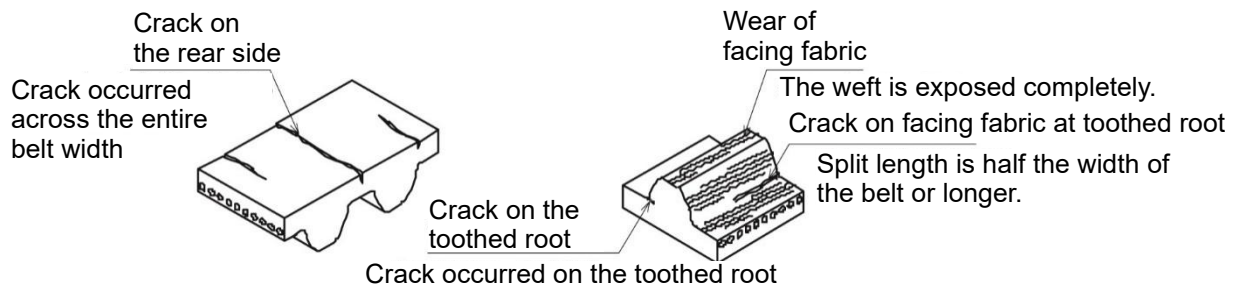


Fig. 5-4

(4) Tension and alignment of reduction belt

- (a) Check that the belt tension is maintained within the range below.

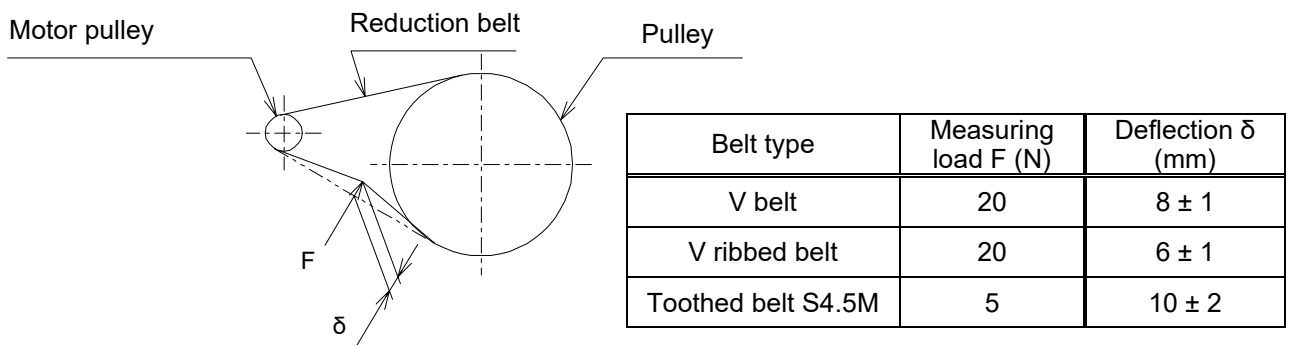
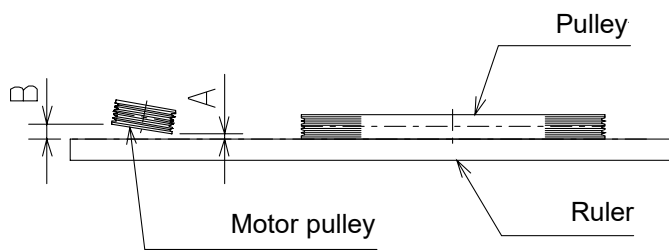


Fig. 5-5

(b) Adjust the alignment of pulleys with the motor mounting bolt so that the belt rotates straight along the pulley.



Belt type	Standard range (mm)
V belt	$A - B \leq \pm 0.35$ $(A + B)/2 = 1 \pm 1.5$
V ribbed belt	$A - B \leq \pm 0.5$
Toothed belt S4.5M	$A \leq 1, B \leq 1$

Fig. 5-6

(5) State of reduction belt

Check the friction power transmission belt (V belt, V ribbed belt) to make sure there is no crack on the areas shown below.

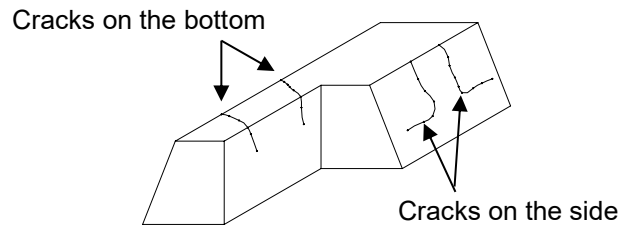


Fig. 5-7

Also, check that there is no partial deformation of the belt as shown below. Since the belt is a chemical product and deteriorates over the years, we recommend replacing the belt in approximately 12 years.

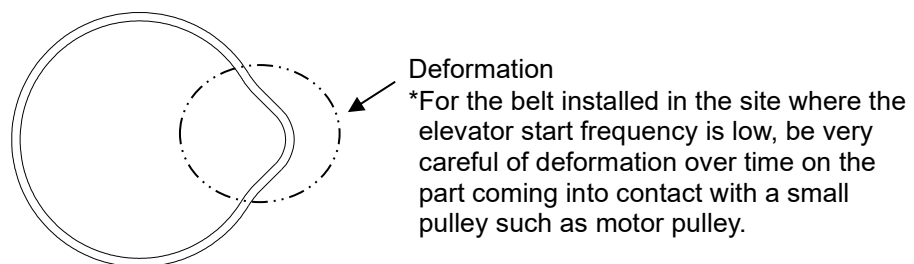


Fig. 5-8

(6) Interlocking chain tension

Check that the interlocking chain tension is maintained within the range shown below. Measure the tension at the center of the chain with the doors fully open.

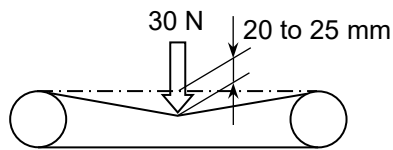


Fig. 5-9

(7) Interlocking rope tension

Check that the interlocking rope tension is maintained within the range shown below. Measure the tension at the center of the rope with the doors fully open.

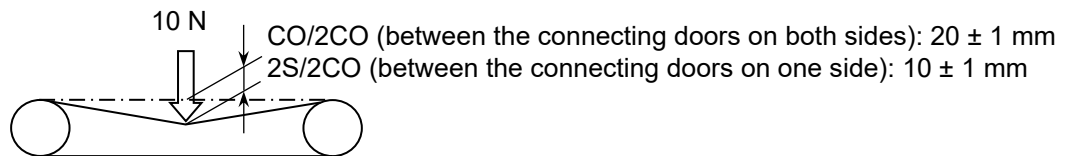


Fig. 5-10

(8) Clearance between upthrust roller and rail

Check that the clearance between the upthrust roller of the door hanger and the door rail is maintained between 0.1 and 0.3 mm.

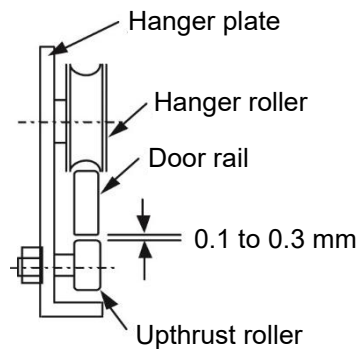


Fig. 5-11

(9) Car-door close detection switch

Check that there is no change in the operating point and installation condition or any other abnormality.

(a) Type 1 (Lever-type forced opening switch)

Check that the switch is activated when the doors move from fully closed position to the position in the opening direction shown in the table below.

Door type	Switch operating point
CO, 2CO (Center-open type)	9 ± 1 mm (vane side) 15 ± 2 mm (side opposite the vane)*
2S, 3S (Side-open type)	9 ± 1 mm

*Only for the belt-driven type with two switches

Check that the switch operation when the doors are fully closed meets the conditions below.

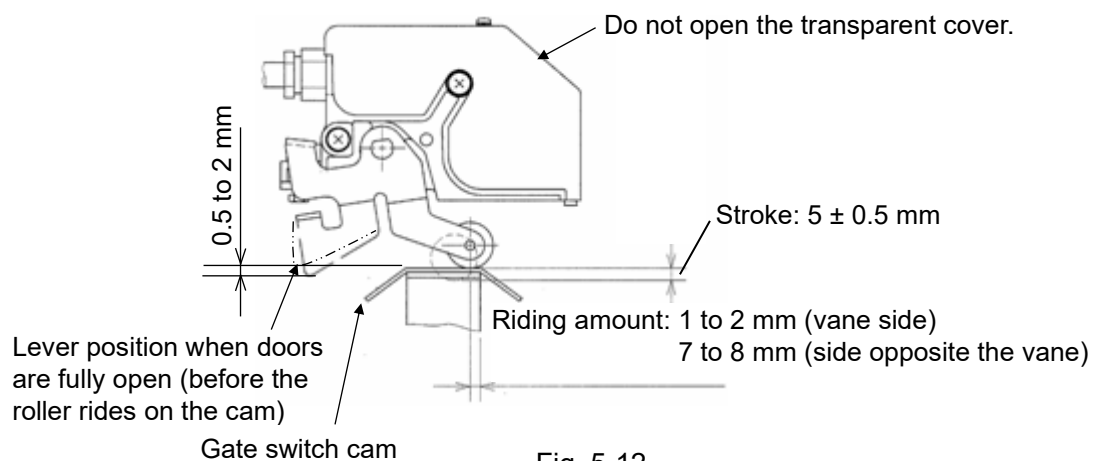
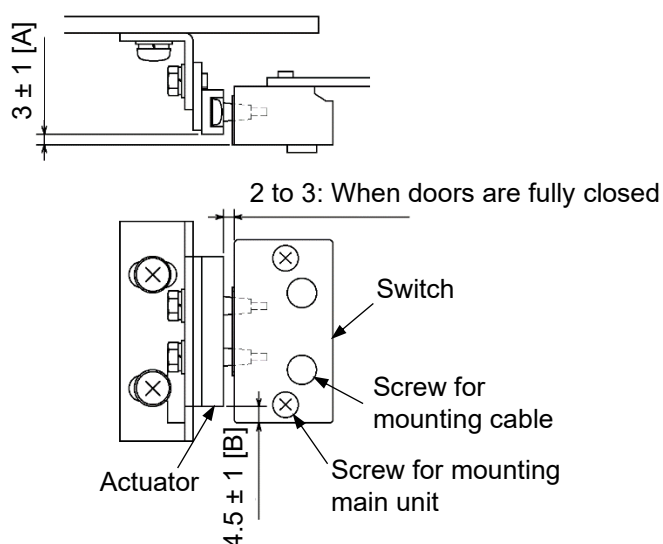


Fig. 5-12

(b) Type 2 (Forced opening switch with separate actuator)

 Warning	
	Shut off the power to the door switch. There is a risk of electric shock if you fasten the screws for mounting cable without shutting off the power.

Check that there is no change in the operating point and installation condition or any other abnormality.



Make sure the total of absolute values of dimensional errors in depth direction [A] and vertical direction [B] is 1.5 mm or less.

Fig. 5-13

Check wear of the actuator.

Check that the length of the actuator is 11.5 mm or more.

Note: Standard tightening torque for screws attached to the switch is as follows.

Screws for mounting main unit (outside): 1.3 to 1.5 N•m

Screws for mounting cable (inside): 0.9 to 1.1 N•m

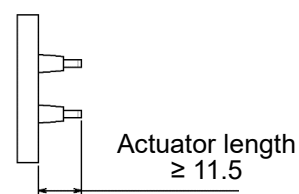


Fig. 5-14

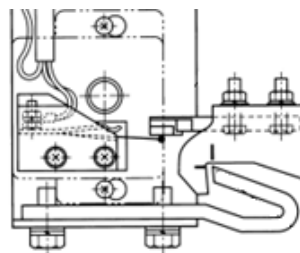
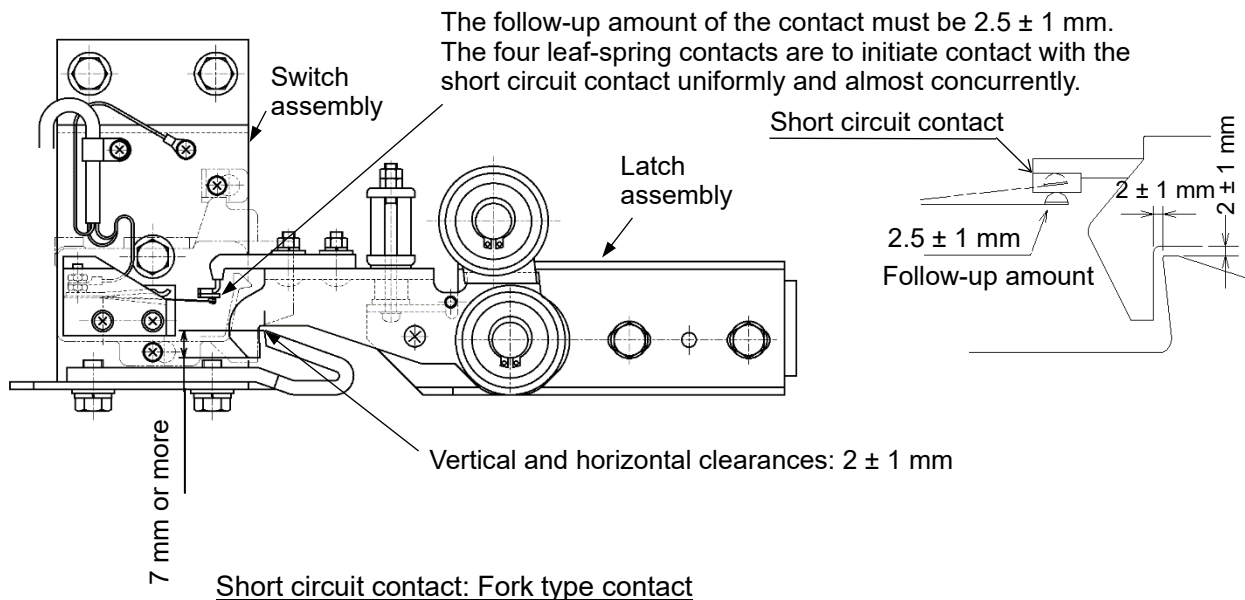
(10) Landing door interlock device

 Warning	
	Shut off the power to the interlock switch. There is a risk of electric shock if you fasten the screws for mounting cable without shutting off the power.

Check that the switch operation when the doors are fully closed is as shown in the figure below.

*Check the type of landing door interlock device by looking at the appearance of the device actually installed.

(a) Type 1 (IL-3X type (*The letter X represents a number.))



Short circuit contact: Flat type contact

(Reference values are the same as those for a fork type contact.)

Fig. 5-15

(b) Type 2 (IL-2X type (*The letter X represents a number.))

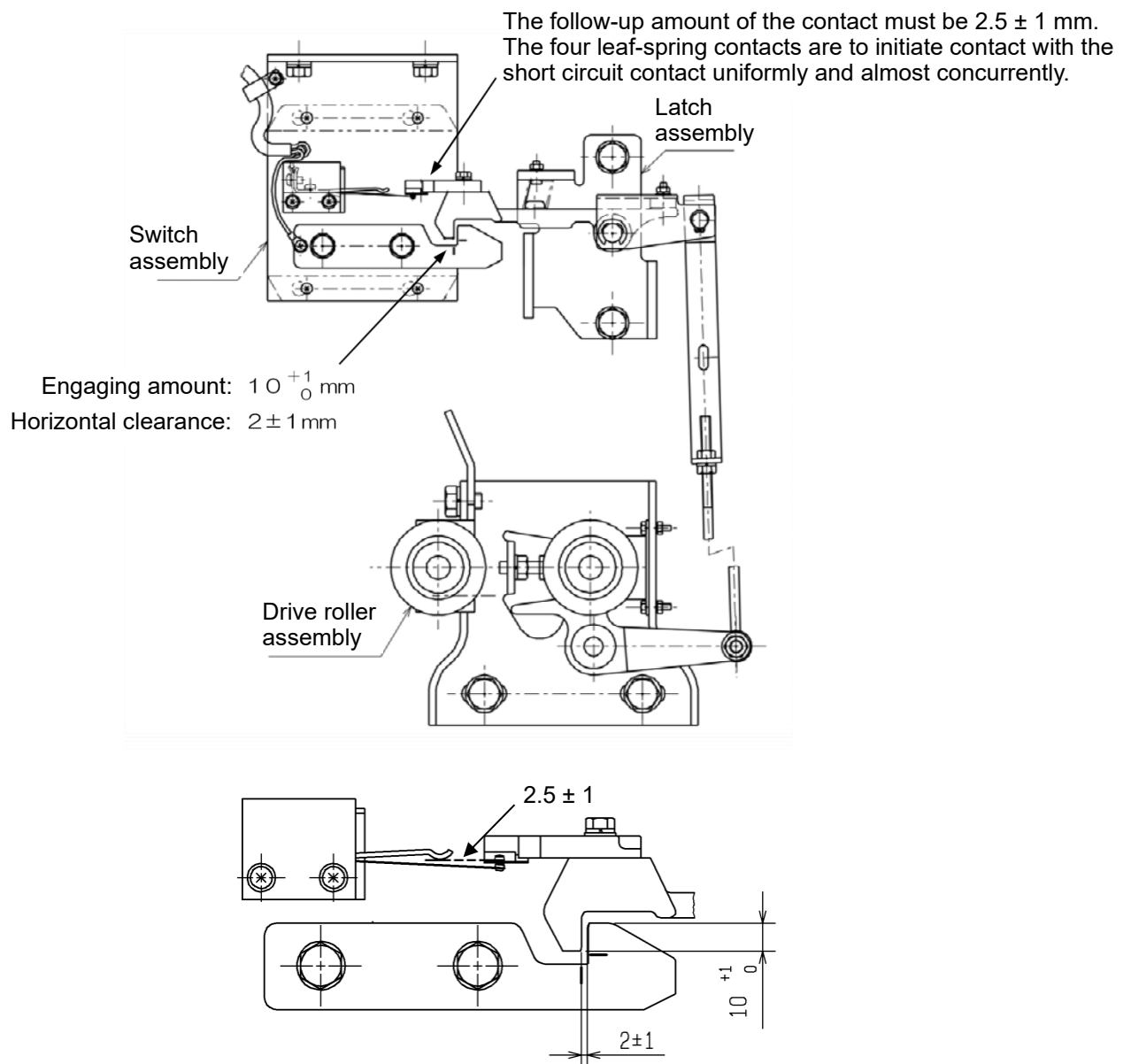


Fig. 5-16

(c) Type 3 (B2 type)

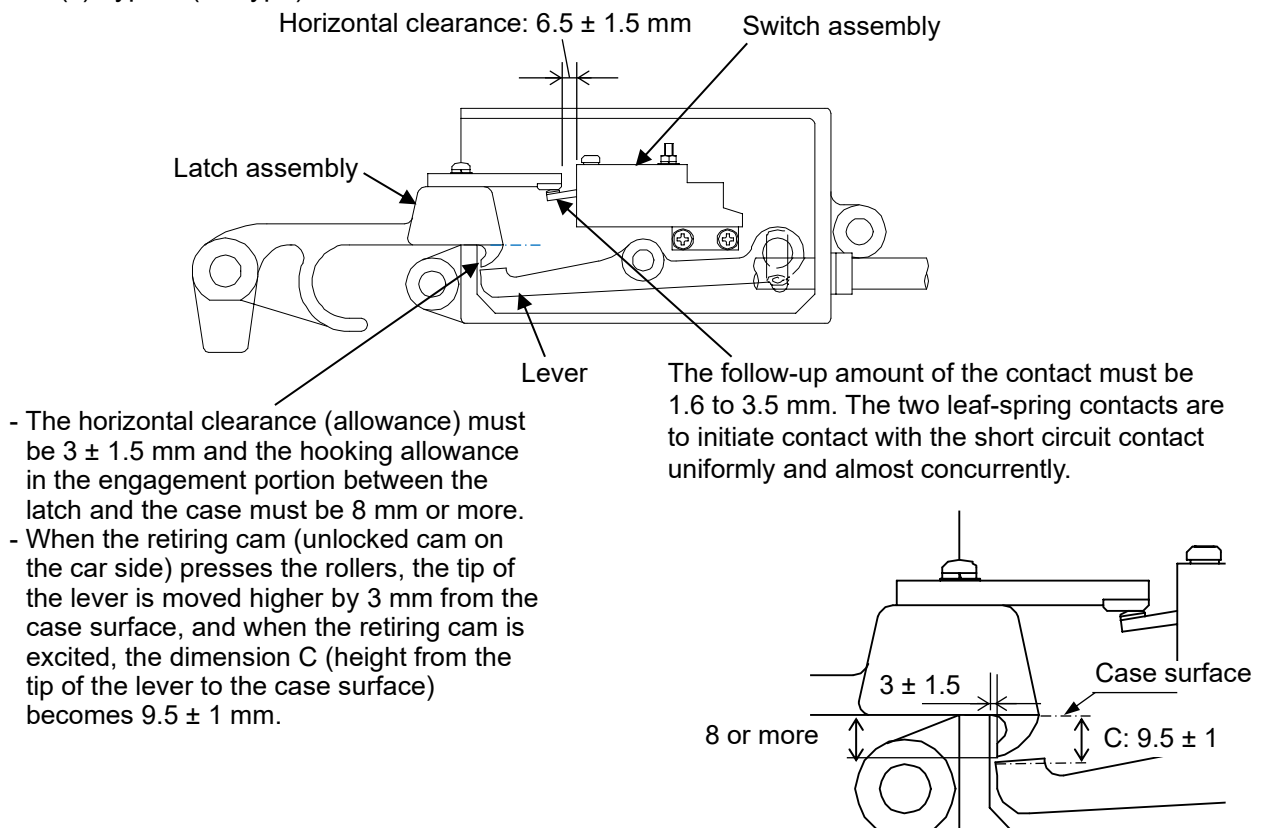


Fig. 5-17

(d) Type 4 (B3 type)

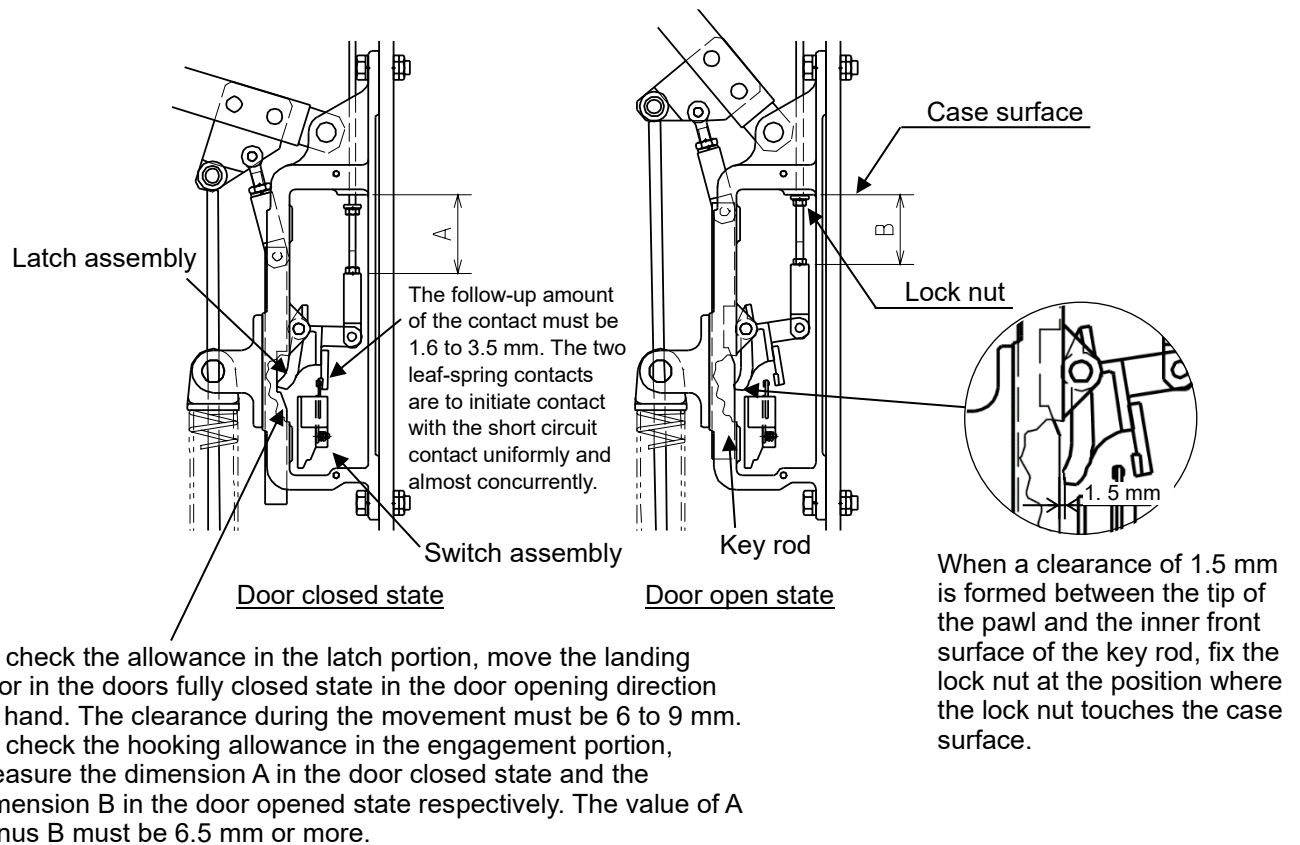
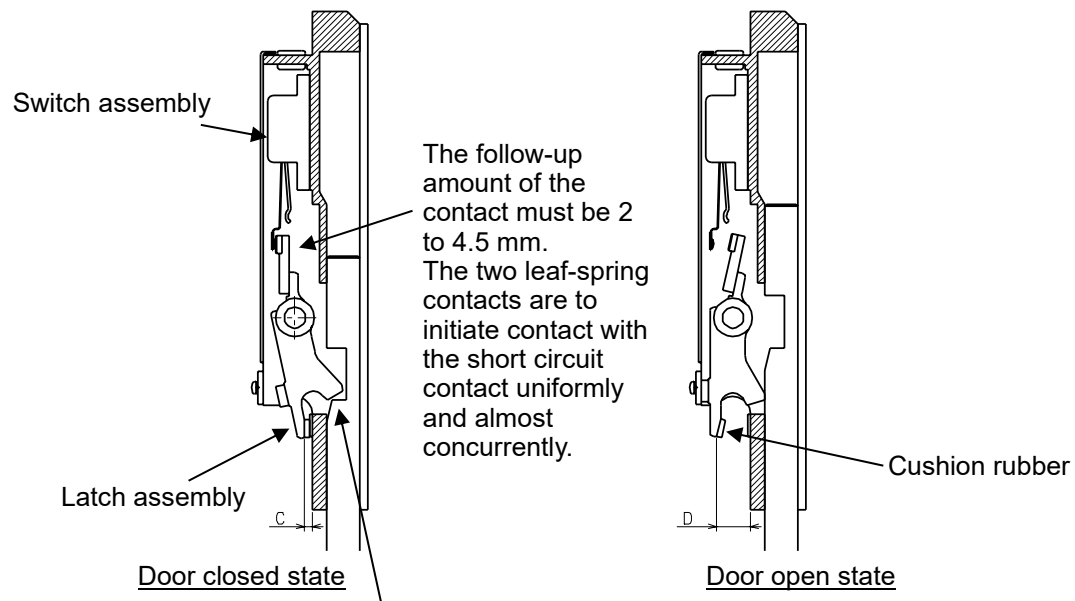


Fig. 5-18

(e) Type 5 (IL-1X type (*The letter X represents a number.))



- To check the allowance in the latch portion, move the landing doors in the door fully closed state in the door opening direction by hand. The clearance during the movement must be 5 to 7 mm.
- To check the hooking allowance in the engagement portion, measure the dimension C in the door closed state and the dimension D in the door opened state respectively. The value of D minus C must be 11 mm or more.

Fig. 5-19

(f) Type 6 (EE-E2 type)

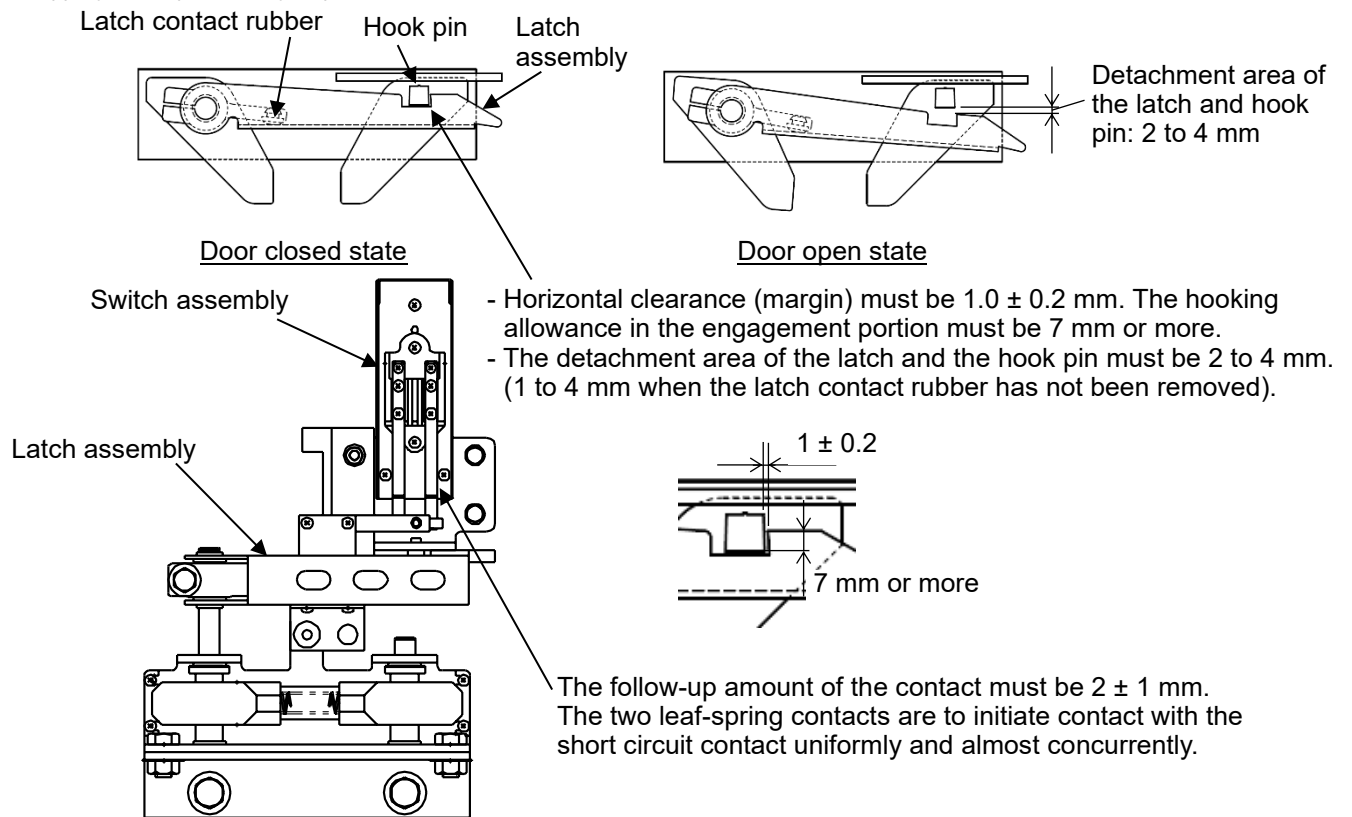
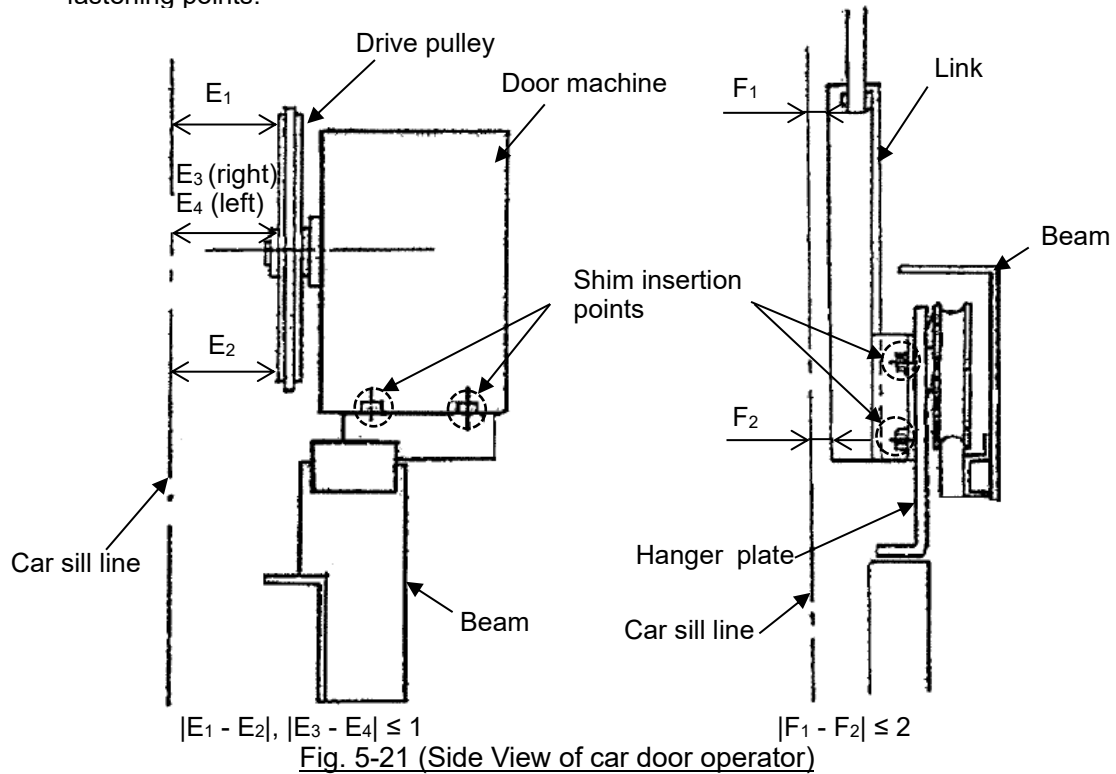


Fig. 5-20

(11) Alignment of link and drive pulley

(a) For double-link driven type

- i) Measure the distances E_1 to E_4 from the car sill line to the drive pulley and the distances F_1 and F_2 from the car sill line to the link.
- ii) Based on the measured values, check that the inclination of the drive pulley in vertical and horizontal directions $E_1 - E_2$, $E_3 - E_4$ and the inclination of the link in vertical direction $F_1 - F_2$ are within the standard range.
- iii) If the inclination is outside the standard range, adjust the position by inserting shims into the fastening points.



(b) For four-link driven type

Check that the clearance between the rear side of link edge and the mounting surface of movable vane on the car door is 27 ± 4 mm. If the clearance is outside the standard range, adjust the inclination of the beam.

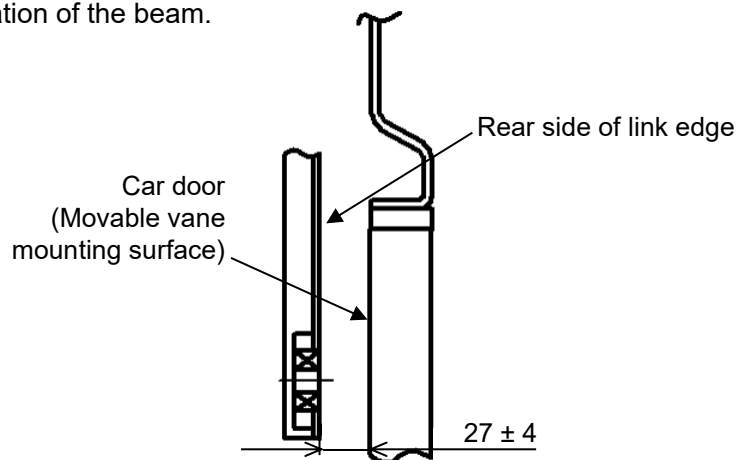


Fig. 5-22

6. Spare parts

To maintain the quality of elevators, we strongly recommend using Mitsubishi's genuine parts.

Note that the spare parts are available for 20 years after the product shipment.

6.1 Long-term repair plans

- To maintain the elevators in legitimate condition over the long term, systematic component replacement is recommended.
- The sample long-term repair plan listed below contains a component replacement plan that is based on the following prerequisites (assumptions). The owner of the installation shall create the actual long-term repair plan and change the plan as necessary by considering the specifications of each elevator, the installation environment and traffic conditions. When asked for advice on maintenance by the owner of the installation, competent maintenance persons shall give appropriate suggestions and advice to the owner of the installation.

Sample long-term repair plan

(Prerequisites)

- Elevators are not exposed to wind, rain, salty wind and hazardous gas and are installed in indoor environments where they are not exposed to direct sunlight.
- The frequency of use (the number of times of start and stop) is approximately 1000 times per day.
- Replacement due to vandalism, misuse and other unavoidable situations, such as earthquakes and lightning strikes, is not included.

Spare parts		Replacement interval (Recommend)
Equipment	Parts	
Car door operator	A set of door hangers	15 to 20 years
	Interlocking rope	12 to 20 years
	Door shoes	5 years
	Belt	12 years
	Pulley	25 years
	Link	25 years
Door safety edge	Connecting cable	14 years
	Switch	14 years
Landing doors	Door shoes for reference floor	6 years
	Door shoes for general floors	9 years
	A set of door hangers	14 to 20 years
	Interlocking rope	14 to 20 years
	Closer pulley (plastic)	10 years
	Interlocking pulley, closer pulley (metal)	25 years
	Link	25 years

*Replacement of the components other than listed above may be required depending on the elevator specifications, installation environment and frequency of use. Also, replacement interval differs depending on the installation environment and frequency of use.